ASTRA P1 SCIENTIFIC RETURN

Scalable output-only acquisition from interacting many-body sources

Observed local noise, finite chronological collection, and a quantum decoder with local controls

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Selected capability. Recover one arbitrary initial many-body quantum state, including its external-reference correlations, using only actually delivered coherent output ports while the source continues to interact. Remove dense synthesis in dimension q^n from this structured acquisition task.

Completed construction. An explicit reverse-history decoder; a uniform operator-support estimate for arbitrary continuous local Hamiltonians in an export-dominated regime; finite local collection from the actual pure Markov ports; an observable weighted stopping certificate; and an exact diagonal-interaction extension at every positive reset rate.

Scope. This is a constructive theorem in a specified calibrated Markov/control interface. It pays finite receiver strength, slew, gate throughput, storage, precision and classical processing. It does not establish a competitive laboratory implementation, a fixed-bandwidth asymptotic device, historical priority, or recovery from arbitrary inaccessible noise.

The previous interaction-order theorem, its meters, the digital causal-cut decoder, and the mixed-output expander companion are preserved without alteration in the accompanying package. They retain their original premises and limits.

Abstract

Consider n source qudits with a known k-local Hamiltonian $H = \sum_X h_X$ and actual coherently delivered pure-vacuum Weyl ports generating independent local depolarization at rate r. The source input is arbitrary and unencoded. Define $\beta = \max_i \sum_{X \ni i} (|X| - 1) \operatorname{osc}(h_X)$. When $\kappa = r - \beta > 0$, an explicit output-only receiver recovers the initial state, uniformly with arbitrary reference, with ideal error at most $2\sqrt{ne^{-\kappa T}}$. The decoder retains reset exports, prepares a maximally mixed dummy source, and reverses the known local Hamiltonian/reset history on receiver registers. Its accuracy follows from an exact projection identity in operator Hilbert space and a support-growth Lyapunov bound. This yields $T = O(\log(n/\varepsilon)/\kappa)$ and $O(nrT)$ expected exported qudits.

A full coherent-dilation comparison, rather than a reduced-generator Trotter argument, permits application of the inherited finite collector only to constant-dimensional local output bins. All receiver controls and compilation costs are polynomial in the declared size, norm, margin and accuracy parameters. Two receiver banks per site can be recycled with explicitly paid endpoint throughput; a separate-bank construction is a conservative alternative. A computable positive-vector certificate bounds individual ideal records under a stronger sufficient rate condition. For Hamiltonians diagonal in a known product basis, revealing each reset's replacement basis state removes quantum reactivation and permits exact heralded acquisition at every $r > 0$. The operation preserves the promised input; it does not compress the whole mixed output stream.

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1 Scientific decision and preserved results

The destination is a quantum interface that makes an interacting experiment available to subsequent quantum computation. Learning a classical description, proving that the reduced source forgets, or establishing the existence of a complementary-channel decoder does not construct that interface.

The previous exact digital decoder records a Boolean active-wire subcircuit and reverses it after all active wires have been exported. Infinitesimal entangling gates defeat its simple branching estimate: Boolean support does not scale with gate strength. We replace that representation by a quantitative operator-support bound tied directly to the error of a physical decoder.

The inherited N1732 theorem collects the full canonical Stinespring channel of a known finite-dimensional Markov source through its actual calibrated pure-vacuum ports, with finite controls and local receiver initialization [1]. Its dense compilation scales in source dimension d. We use it only for $d = q$, never $d = q^n$. The whole-bath mixed-output theorem has separate premises [2]; the preserved expander companion warns against inferring cheap whole-output storage from a small source description. The present target is the initial payload and its reference correlations, not every correlation in the emitted bath.

The completed weak-control interaction-order theorem and hidden-dephasing acquisition law are unchanged. Accessible depolarizing exports are different physical resources from inaccessible Z noise. The present ideal theorem has no additional hidden source noise; Section 10 prices perturbations explicitly.

2 Source, ports, receiver and operational promise

The source is $S = (\mathbb{C}^q)^{\otimes n}$, with arbitrary initial state ρ_{SR} and an arbitrary inaccessible reference R. It starts unencoded and independent of the receiver. The receiver may prepare protected local ancillas and use known finite-dimensional local quantum controls. It never acts on the surviving source, feeds back an unaccounted field, or accesses lost outputs.

For the primary finite compilation statement take a known static Hamiltonian

$$H = \sum_{X \in \mathcal{E}} h_X, \qquad |X| \le k, \qquad m = |\mathcal{E}|. \tag{1}$$

The matrices h_X and their supports are explicitly supplied. For scalable families, q, k are fixed, m is polynomial, and matrix precision and norm scales are charged. A time-dependent extension requires an efficiently computable description and a finite simulation budget; arbitrary real-function oracles are not efficient inputs.

Let $W_0 = I, W_1, \dots, W_{q^2-1}$ be an orthogonal Weyl unitary basis. The actual local vacuum ports are

$$L_{i\mu} = \frac{\sqrt{r}}{q} W_{\mu,i}, \qquad \mu \neq 0. \tag{2}$$

Their labels and coherent phases are calibrated, and all outputs are delivered coherently and losslessly to the receiver. They generate

$$\mathcal{L}(\rho) = -i[H, \rho] + r \sum_i (\mathcal{R}_i - \mathrm{id})(\rho), \qquad \mathcal{R}_i(\rho) = \frac{I_i}{q} \otimes \mathrm{Tr}_i\, \rho. \tag{3}$$

The physical photon rate at each site is $c = r(1 - q^{-2})$, not r. A reduced depolarizing generator alone does not supply these ports.

Define the computable local bounds

$$\beta = \max_i \sum_{X \ni i} (|X| - 1)\,\mathrm{osc}(h_X), \qquad \kappa = r - \beta, \tag{4}$$

$$\chi = \max_i \sum_{X \ni i} \mathrm{osc}(h_X), \qquad \mathrm{osc}(h) = \lambda_{\max}(h) - \lambda_{\min}(h). \tag{5}$$

One-site fields contribute to control, timing and simulation costs but not to β. These bounds depend on the supplied local decomposition; cancellations in the total Hamiltonian do not justify replacing them by a total-norm estimate.

The main output is an n-qudit receiver register S' with

$$\sup_{\rho_{SR}} \|(\mathcal{A} \otimes \mathrm{id}_R)(\rho_{SR}) - \rho_{S'R}\|_1 \le \varepsilon. \tag{6}$$

The supremum may include all reference dimensions. One copy of the unknown input is consumed. There is no fresh-copy distillation or retry of a destroyed state.

3 The acquired reset/export operation

For a noise-only bin of duration Δ, the local reduced channel is

$$\mathcal{D}_p(\rho) = (1-p)\rho + p\,\mathrm{Tr}(\rho) I/q, \qquad p = 1 - e^{-r\Delta}. \tag{7}$$

After finite chronological collection of its actual output into a constant-dimensional receiver, a known local Kraus isometry gives

$$V|\psi\rangle = \sqrt{1-p}\,|\psi\rangle_S|\emptyset\rangle + \sqrt{p}\,|\Phi_q\rangle_{SA}|\psi\rangle_B. \tag{8}$$

The two flag sectors are orthogonal. Reading only the flag and discarding A produces

$$\mathcal{I}_0(\rho_{SR}) = (1-p)\rho_{SR}, \qquad \mathcal{I}_1(\rho_{SR}) = p\frac{I_S}{q} \otimes \rho_{BR}. \tag{9}$$

The old quantum state, including entanglement with the rest of the experiment, is exported. The flag is input independent. The internal payload label is not measured. This is a synthetic Kraus label produced after coherent acquisition, not a supposed direct photon-count record.

For an exact compilation target, write $a = e^{-r\Delta}$, $p_0 = [1 + (q^2-1)a]/q^2$, $p_1 = (1-a)/q^2$. The minimal Kraus operators may be chosen as $K_0 = \sqrt{p_0} I$, $K_\mu = \sqrt{p_1} W_\mu$. Rows $\emptyset, (a,b)$ and columns μ of the converting isometry are

$$u_{\emptyset,0} = \sqrt{a/p_0}, \qquad u_{\emptyset,\mu\neq 0} = 0, \tag{10}$$

$$u_{ab,0} = \frac{\sqrt{1-a}\,\delta_{ab}}{q\sqrt{qp_0}}, \qquad u_{ab,\mu\neq 0} = \frac{\overline{(W_\mu)_{ab}}}{\sqrt{q}}. \tag{11}$$

Weyl orthogonality gives $u^\dagger u = I$ and reset Kraus operators $\sqrt{(1-a)/q}\,|a\rangle\langle b|$. The apparent q^2+1 environment dimension costs only a constant local extension at fixed q.

An ideal continuous register model uses independent rate-r reset/export clocks. It can also be represented by adding a calibrated source-independent coherent identity port and changing Weyl marks into matrix-unit marks. The finite construction below does not require that extra port: it uses the actual nonidentity-Weyl outputs and the finite-bin conversion above.

4 The explicit decoder and its exact error identity

Condition on an ideal reset record γ with M exports. Each reset is represented mathematically by a SWAP with a fresh maximally mixed bath qudit. This representation describes the acquired instrument; it grants no source SWAP control. The resulting known unitary U_γ consists of source Hamiltonian segments and these wire exchanges. With $d = q^n$, $b = q^M$, its export channel is

$$\mathcal{C}_\gamma(\rho) = \mathrm{Tr}_S\, U_\gamma(\rho \otimes I_B/b)U_\gamma^\dagger. \tag{12}$$

Prepare an independent dummy source S' in I/d and apply

$$\boxed{\mathcal{D}_\gamma(\sigma) = \mathrm{Tr}_B\, U_\gamma^\dagger(I_{S'}/d \otimes \sigma)U_\gamma.} \tag{13}$$

This is a deterministic quantum channel. In the receiver, reverse the last Hamiltonian interval, SWAP the indicated dummy site with the last export, discard the consumed export, and continue in reverse record order. Return the dummy register. The physical field is never reversed and the surviving source is never accessed.

Because $\mathcal{C}_\gamma(I/d) = I/b$, this is the Petz channel $(b/d)\mathcal{C}_\gamma^\dagger$ at the maximally mixed prior. The scalar is not a postselection probability: Equation (13) is its direct trace-preserving implementation. Such inverse-dilation/Petz connections are established machinery [3, 4, 5].

Use normalized Hilbert–Schmidt inner products. The operator-space embedding

$$\mathcal{J}_\gamma(X) = U_\gamma(X \otimes I_B)U_\gamma^\dagger \tag{14}$$

is an isometry. Let P_0 be conditional expectation onto source-identity operators:

$$P_0(O) = I_S \otimes \mathrm{Tr}_S(O)/d. \tag{15}$$

Direct substitution gives the exact identity

$$\boxed{\begin{aligned} \mathcal{D}_\gamma\mathcal{C}_\gamma &= \mathcal{J}_\gamma^\dagger P_0 \mathcal{J}_\gamma, \\ \langle X, (\mathrm{id} - \mathcal{D}_\gamma\mathcal{C}_\gamma)X\rangle &= \|(I - P_0)\mathcal{J}_\gamma(X)\|_2^2 . \end{aligned}} \tag{16}$$

Thus $0 \le \mathcal{D}_\gamma\mathcal{C}_\gamma \le I$ as operators on operator Hilbert space. This does *not* say that $\mathrm{id} - \mathcal{D}_\gamma\mathcal{C}_\gamma$ is a completely positive physical map.

For each source site let P_i project onto its identity operator and $Q_i = I - P_i$. The commuting projections define

$$W = \sum_i Q_i, \qquad I - P_0 \le W \le nI. \tag{17}$$

This observable counts operator support still depending on the source. Support already exported into B is harmless. The proof therefore controls the error of the actual decoder, not merely the source's reduced-state mixing.

5 Continuous interactions and the acquisition theorem

Let $O(t)$ be an initial operator evolved through the conditional source/bath unitary. Its normalized Hilbert–Schmidt norm is constant. A reset at i sets $\langle O, Q_i O\rangle$ to zero and leaves the other terms unchanged. Independent clocks give an exact average drift $-r\langle W\rangle$.

For a term h_X with $s = |X|$, the self-adjoint operator $K_X = \mathrm{ad}_{h_X}$ on operator Hilbert space has norm $\mathrm{osc}(h_X)$. It annihilates the local identity sector and preserves its complement

$$P_X = I - \prod_{i \in X} P_i. \tag{18}$$

On that complement $W_X = \sum_{i\in X} Q_i$ has spectrum in $\{1,\dots,s\}$. Hence

$$\left\|W_X - \tfrac{s+1}{2}P_X\right\| \le \tfrac{s-1}{2}, \qquad i[K_X, W_X] \le (s-1)\operatorname{osc}(h_X)P_X \le (s-1)\operatorname{osc}(h_X)W_X. \tag{19}$$

Summing the terms yields Hamiltonian growth at most $\beta\langle W\rangle$. Therefore

$$\frac{d}{dt}\mathbb{E}\langle O(t), WO(t)\rangle \le -(r-\beta)\mathbb{E}\langle O(t), WO(t)\rangle. \tag{20}$$

Writing $\overline{\mathcal{R}}_T = \mathbb{E}_\gamma \mathcal{D}_\gamma \mathcal{C}_\gamma$, Equation (16) gives the uniform operator inequality

$$0 \le I - \overline{\mathcal{R}}_T \le \delta_T I, \qquad \delta_T = \min\{1, ne^{-\kappa T}\}. \tag{21}$$

This order tensors with the identity on any reference's operator Hilbert space. For every normalized pure joint input Ψ,

$$\operatorname{Tr}\Big[\Psi(\overline{\mathcal{R}}_T \otimes \mathrm{id})(\Psi)\Big] \ge 1 - \delta_T. \tag{22}$$

Pure-state fidelity and trace distance therefore imply the dimension-free result below.

Theorem 1 (Output-only recovery under local export)**.** *For the ideal acquired reset/export process and a known continuous local Hamiltonian, if $\kappa = r - \beta > 0$, the explicit decoder (13) satisfies*

$$\boxed{\left\|\overline{\mathcal{R}}_T - \mathrm{id}\right\|_\diamond \le 2\sqrt{ne^{-\kappa T}}.} \tag{23}$$

The full classical record can be retained with the same bound relative to an input-independent record tensored with the original payload. The mean number of exported qudits is nrT. A logarithmic collection time in n/ε suffices at fixed positive κ.

For the full-record assertion, fix a pure reference input and let f_γ be its conditional recovered overlap. Equation (21) gives $\mathbb{E}(1-f_\gamma) \le \delta_T$. The trace norm of the classical-record block diagonal output is the weighted sum of block trace norms, bounded by $2\mathbb{E}\sqrt{1-f_\gamma} \le 2\sqrt{\delta_T}$. This does not assume an unjustified average of worst-record diamond errors.

Full general consequence. Compose the acquired channel with any subsequent finite quantum procedure. Channel contraction adds at most ε to its error, with only the stated polynomial acquisition overhead. Under these premises, coherent output-only access supplies the initial quantum register for any subsequent experiment with polynomial overhead. This includes reference-sensitive and collective measurements. It does not grant a new input-loading operation or a communication rate without the associated preparation and latency costs.

6 Finite chronological acquisition of the real continuous output

The theorem above is not used as a license to assume quantum jump registers. We now construct them from the actual pure Weyl outputs.

6.1 A bound on the full coherent dilation

Divide $[0,T]$ into K bins of width $\Delta = T/K$. Compare the actual continuous evolution with a fictitious process that has noise-only local evolution within each bin and the exact Hamiltonian bin unitary at its end. These kicks are a comparison device, not source controls performed by the receiver.

The actual nonidentity Weyl ports have scalar total intensity $R_{\mathrm{ph}} = nc \le nr$. Conditional on coarse time/site histories their internal quantum-mark maps are isometries. A jump at site i intertwines every Hamiltonian term disjoint from i. Centering touching terms gives

$$\|H_{\mathrm{out}} J_i - J_i H_{\mathrm{in}}\| \le \chi. \tag{24}$$

Moving Hamiltonian evolution within a bin past one event costs at most $\chi\Delta$. For a full history with M events the isometry discrepancy is at most $\chi\Delta M$. Orthogonal full timing sectors and the Poisson second moment give

$$\boxed{\eta_{\mathrm{split}} \le 2\chi\Delta\sqrt{R_{\mathrm{ph}}T + (R_{\mathrm{ph}}T)^2}.} \tag{25}$$

A more conservative per-bin triangle bound is $2\chi T\sqrt{R_{\mathrm{ph}}\Delta + (R_{\mathrm{ph}}\Delta)^2}$. Either controls the *entire source-plus-output channel*, with arbitrary reference. A reduced-source generator estimate would not be enough.

The finite collector is the same physical receiver on the actual and comparison outputs. Postprocessing contraction preserves (25). There is no Boolean activation of an infinitesimal interaction.

6.2 Local finite receiver and exact control formulas

Each noise-only local bin has source dimension q. Invoke N1732 only at that dimension. The local guide has a particularly explicit form. Let $P = \Phi_q$, $Q = I - P$, regularization $e \in (0, 1)$, and

$$a(t) = (1-e)e^{-rt}, \qquad \lambda_0(t) = \frac{1 + (q^2-1)a(t)}{q^2}, \qquad \lambda_1(t) = \frac{1-a(t)}{q^2}. \tag{26}$$

The guides commute, so the gauge is the identity and the receiver Hamiltonian is zero. For $A_\mu = I \otimes L_\mu$, the actual receiver couplings are

$$C_\mu(t) = -\left[QA_\mu Q + \sqrt{\lambda_1/\lambda_0}\, PA_\mu Q + \sqrt{\lambda_0/\lambda_1}\, QA_\mu P\right]^T. \tag{27}$$

Only two scalar waveforms multiply fixed matrices. The q^2-level receiver starts in a Bell state between its own two q-level subsystems, independent of source and reference. The known endpoint contraction is diagonal in P, Q; its positive completion is deterministic. The regularization contribution to full channel error is at most $2\sqrt{2e} + 2e$.

For a total local collection-plus-instrument error $\eta \le 1$, a sufficient budget is

$$e = \eta^2/256, \qquad m_{\mathrm{col}} = \max\left\{1, \left\lceil \frac{88c^2q^4\Delta^2}{\eta e^2} \right\rceil\right\}. \tag{28}$$

Finite ramps of length

$$\tau = \min\{\Delta/4, \eta e/(160cq^2)\} \tag{29}$$

make the coupling vanish at both endpoints. Peak and slew obey

$$\max_\mu \|C_\mu\| \le 16\sqrt{r}/\eta, \tag{30}$$

$$\max_\mu \left\|\dot{C}_\mu^{\mathrm{finite}}\right\| = O_q\left(\frac{\sqrt{r}}{\eta\Delta} + \frac{r^{3/2}}{\eta^4}\right), \tag{31}$$

$$m_{\mathrm{col}} = O_q(1 + r^2\Delta^2\eta^{-5}). \tag{32}$$

This pays finite coefficient strength and slew; it is not a claim of fixed microscopic bandwidth. The complete local audit gives the ramp and calibration estimates. In particular, for coefficient errors at most $\xi \le M_C = \sqrt{r/e}$,

$$\eta_{\mathrm{cal}} \le \Delta(q^2-1)(6\sqrt{r}/q + 6M_C)\xi + 2\Delta \left\|\delta H_M\right\|. \tag{33}$$

Inverse-polynomial precision suffices. Calibration must include actual channel phases and labels, not just a fitted reduced generator.

There are $B = nK$ local bin uses. Choosing $\eta = \eta_{\mathrm{collect}}/B$ bounds the total instrument error by η_{collect}. Constant-dimensional endpoint gates convert the captured canonical environment to (8). No exact physical photon-time detector or global field unitary is assumed.

6.3 Why the analysis remains uniform under refinement

The comparison bin has a local reset layer with $p = 1 - e^{-r\Delta}$ followed by a Hamiltonian unitary. Expected operator support shrinks by $e^{-r\Delta}$ in the reset layer and grows by at most $e^{\beta\Delta}$ under the unitary. Therefore each bin multiplies the upper bound by at most $e^{-\kappa\Delta}$, independently of how small Δ is. The same decoder theorem applies. It reverses the known bin unitaries and the acquired reset records on receiver memory.

7 An explicit finite resource bill

Let $0 < \varepsilon \le 1$, $\kappa > 0$, and use $R_+ = nr$. One conservative complete allocation is

$$T = \kappa^{-1}\log(400n/\varepsilon^2), \tag{34}$$

$$K = \max\left\{1, \left\lceil \frac{20\chi T\sqrt{R_+T + (R_+T)^2}}{\varepsilon} \right\rceil \right\}, \qquad \Delta = T/K, \tag{35}$$

$$B = nK, \qquad \eta = \varepsilon/(20B). \tag{36}$$

The ideal decoding and coherent splitting contributions are each at most $\varepsilon/10$. The collector at each use has error at most η; give its associated queue, readout and refresh cycle a separate error allowance at most η. Their summed collection-plus-cycling error is at most $2B\eta = \varepsilon/10$. Allocate at most $\varepsilon/10$ each to finite memory-cap failure, Hamiltonian simulation, additional decoder gates, protected-memory waiting, calibrated model error, and any separately bounded loss. Their sum is at most 0.9ε, leaving slack. These are sufficient parameters, not optimized device settings.

7.1 Finite memory and chronological scheduling

The ideal binned export count has mean $nK(1-e^{-r\Delta}) \le nrT$. For each site, its count has moment generating function at one bounded by $\exp[(e^1-1)rT]$. With a per-site queue capacity

$$m_* = \left\lceil (e^1-1)rT + \log(20n/\varepsilon) \right\rceil, \tag{37}$$

a union bound makes any queue overflow have probability at most $\varepsilon/20$. Replacing the overflow output by a fixed state adds at most $\varepsilon/10$ in diamond norm.

A conservative interface can allocate one collector bank per site/bin: $O(B)$ quantum workspace, finite endpoint gates after collection, and no instantaneous recycling assumption. A reusable implementation instead has two fixed cascaded banks per site, alternating active bins. Inactive couplings are zero, and the known propagation delays shift their control windows. Give each bank its own local queue of capacity m_*. This removes contention and propagation-induced ordering

problems. An individual queue overflow implies that the site's total count exceeded m_*, so the same failure bound applies.

During its inactive bin, a bank performs its endpoint conversion, queues the payload export, reads the flag and reinitializes its Bell state. These operations must satisfy

$$\tau_{\text{endpoint}} + \tau_{\text{queue}} + \tau_{\text{readout}} + \tau_{\text{refresh}} \le \Delta. \tag{38}$$

With a finite operation count g_{end} per endpoint, it requires the explicitly paid full-operation rate

$$\Omega_{\text{end}} = \Omega(g_{\text{end}}/\Delta). \tag{39}$$

Nearest-neighbour insertion into a local queue costs at most $O(m_*)$ swaps. Give individual queue gates error $O(\eta/(m_*+1))$ and fixed-dimensional endpoint circuits their assigned total error. Thus g_{end} and Ω_{end} are polynomial. If the quoted clock counts only unitary gates, separate measurement and refresh latencies are required.

For an explicit finite refresh, isolate the bank from the main field and use auxiliary receiver reset ports $L_j^{\text{reset}} = \sqrt{\Gamma}|0\rangle\langle j|$, $j \neq 0$, on its fixed-dimensional register. Pure-reset channel error is at most $2\sqrt{2}e^{-\Gamma\tau_{\text{refresh}}/2}$. Therefore

$$\Gamma \ge \frac{2}{\tau_{\text{refresh}}} \log \frac{2\sqrt{2}}{\eta_{\text{refresh}}} \tag{40}$$

suffices, followed by finite local Bell preparation. The main traveling-field coupling stays zero throughout refresh. This is an additional paid receiver resource, not new source access.

This schedule gives total quantum memory $O(n(m_*+1))$, including two queues and two collectors per site, the dummy source and ancillary workspace. If a hardware platform cannot meet the endpoint deadline, use more banks and count them. The actual flag law need not remain exactly binomial: writing $\nu_{\text{front}} = \eta_{\text{split}} + \eta_{\text{collect+cycle}}$, actual overflow probability is at most $\varepsilon/20 + \nu_{\text{front}}/2$. Comparing the actual capped protocol with the ideal capped and then uncapped protocols adds only $\nu_{\text{front}} + \varepsilon/10$ to the recovery error.

Sparse reset records use $O(nm_* \log(nK))$ classical bits. Every local flag must still be processed: the interface work is $B = nK$, not just the number of exported qudits. At fixed q, k, r, χ, κ, $T = O(\log(n/\varepsilon))$, $B = O(n^2T^2/\varepsilon + n)$, and reusable quantum memory is $O(n \log(n/\varepsilon))$.

Total programmed collector intervals are

$$O_q\Big(B[1 + r^2\Delta^2\eta^{-5}]\Big). \tag{41}$$

The waveform can be reused, but its executions and control updates remain real work. Near-linear retained memory and logarithmic collection time do not mean near-linear total physical control cost.

7.2 Inverse dynamics, gates and waiting

Let $C_H = \sum_{a<b} \|[h_a, h_b]\|$. Consecutive bin unitaries with no intervening reset can be combined. For $C_H > 0$, a first-order product formula with internal step $s = O(\varepsilon_{\text{sim}}/(C_H T))$ yields a local-gate count

$$G_H = O\left(m\left[\frac{C_H T^2}{\varepsilon_{\text{sim}}} + nm_* + 1\right]\right). \tag{42}$$

For $C_H = 0$, the commuting product is exact on each interval and the same gate bound holds with its first term zero. There are at most $2nm_*$ reverse swaps with the two-queue schedule. Fixed q, k make each local exponential a constant-dimensional synthesis problem. A conservative nearest-neighbour routing scheme adds polynomial swaps; no exponentially large gate is primitive.

Known analog replay of $-H$ is an alternative only when that receiver interaction is actually supplied.

Each of the actual G compiled operations can be given diamond error at most $\varepsilon_{\mathrm{gate}}/G$. Storage must last for collection time, propagation delay, endpoint processing and the decoder runtime. If Q_{mem} qudits are provisioned for a total wall time T_{wall}, a per-qudit error-generator bound ν_{mem} contributes at most $Q_{\mathrm{mem}} T_{\mathrm{wall}} \nu_{\mathrm{mem}}$. This quantity must fit the allocated storage budget. The theorem can use a separately specified fault-tolerant receiver, but does not hide its overhead or grant fixed physical error per operation for arbitrarily large computations.

All costs are polynomial in n, m, q^k, the explicit local norm/time scales, κ^{-1} and ε^{-1}, with standard precision and routing factors. This is not polynomial in the bit length of an exponentially large physical coupling at no additional cost; no fast-forwarding claim is made.

8 Observable weighted stopping

The stronger fixed-time theorem needs $r > \beta$. A slightly more conservative, classically computable certificate is also available. Let $v_i(0) = 1$ and, between reset events, evolve

$$\dot{v}_i = \sum_{\substack{X \ni i \\ |X| \geq 2}} \mathrm{osc}(h_X) \sum_{j \in X} v_j. \tag{43}$$

At an observed reset set $v_i \leftarrow 0$. For any initial operator, the actual source-support components $x_i = \|Q_i O\|_2^2$ obey the same differential upper bounds, because

$$\|[\mathrm{ad}_{h_X}, Q_i]\| \leq \mathrm{osc}(h_X) \tag{44}$$

on the nonidentity sector. Positive-system comparison gives $x_i \leq v_i \, \|O(0)\|_2^2$. Values can safely be clamped to one. Thus an ideal record γ certifies

$$\boxed{\|\mathcal{D}_\gamma \mathcal{C}_\gamma - \mathrm{id}\|_\diamond \leq 2\sqrt{\min\{1, \sum_i v_i\}}.} \tag{45}$$

The computation uses only known interaction bounds and observed reset flags. It does not measure the payload.

Define $\beta_{\mathrm{cert}} = \max_j \sum_{X \ni j, \, |X| \geq 2} |X| \, \mathrm{osc}(h_X)$ and $\kappa_{\mathrm{cert}} = r - \beta_{\mathrm{cert}}$. If $\kappa_{\mathrm{cert}} > 0$, the first time τ_a at which $\sum_i v_i \leq a$ satisfies

$$\Pr(\tau_a > t) \leq \min\{1, (n/a) e^{-\kappa_{\mathrm{cert}} t}\}, \qquad \mathbb{E}\tau_a \leq \frac{\log(n/a) + 1}{\kappa_{\mathrm{cert}}}. \tag{46}$$

A finite cutoff adds a declared failure probability. In the binned version apply the positive matrix exponential and reset masks in the actual comparison order. Certified upper rounding or interval arithmetic is required; this n-variable calculation is polynomial.

The per-record guarantee is exact for the ideal acquired instrument. A finite collector's unconditional diamond error does not give the same conditional bound on every rare observed record. For a capped accepted map of ideal success s, physical instrument error η_{phys} must be divided by a lower bound on the actual success probability when reporting conditional accuracy. The main fixed-time unconditional theorem avoids that issue. A physical stopping-time saving additionally requires the paid endpoint and certificate-processing latency; deferred flags in the all-bank construction can identify a decodable earlier prefix afterwards, but cannot shorten collection retroactively.

9 Exact diagonal interactions at every positive export rate

Suppose $H(t)$ is diagonal in a known fixed product basis, with a supplied efficiently integrable local description. At every reset, measure the purifier A in that basis. Equation (8) shows that this prepares the fresh source site in a uniformly random *known* basis state, independently of the old payload exported in B.

Such a site remains in a known basis state under diagonal interactions. It can impose a known phase on unreset quantum sites but cannot become a new quantum carrier. Retain only the first B export from each source site and record all subsequent replacement labels until collection ends.

For an initial computational basis string x, define $z_i(t;x) = x_i$ before that site's first reset and equal to its recorded fresh basis label thereafter. When all sites have reset, the retained payload is

$$|x\rangle \longmapsto e^{-if_\gamma(x)}|x\rangle, \qquad f_\gamma(x) = \int_0^T \sum_X h_X(t; z_X(t;x))\, dt. \tag{47}$$

The remaining source and discarded exports have no input dependence. Inverting this known phase recovers every superposition and reference exactly on the completed ideal records. Substituting classical labels leaves a sum of at-most-k-local diagonal functions. Accumulate one table $F_{X,\gamma}(x_X)$ per Hamiltonian term, so that

$$f_\gamma(x) = \sum_X F_{X,\gamma}(x_X), \qquad U_{\mathrm{corr}} = \prod_X e^{iF_{X,\gamma}(B_X)}. \tag{48}$$

All correction gates commute. There is only one final fixed-dimensional diagonal gate per Hamiltonian term. With maximum term degree D and M reset flags, the classical integration and table work is $O(q^k[m + DM])$ and table storage is $O(mq^k)$ numbers at the declared finite precision. Streaming stores the n first exports and $O(n)$ collector workspace; later exports and the full classical history need not be retained. Every one of the nK local interface executions and its finite cycling deadline remains charged.

For independent rate-r clocks, or the matching binned reset instrument,

$$\Pr[\text{a site has not reset by } T] \le ne^{-rT}. \tag{49}$$

Thus $T = \log(2n/\varepsilon)/r$ suffices for unconditional ideal error at most ε, with exactly n retained payload qudits. Interaction strength affects phase precision, timing and finite interface costs but not this completion probability. This family proves that $r > \beta$ is a constructive sufficient regime, not a universal threshold. General noncommuting weak-export Hamiltonians remain unresolved.

10 Errors, physical limits and preserved boundaries

Hidden source noise. If an additional inaccessible generator has diamond norm at most ν_{hid}, a Duhamel bound adds at most $T\nu_{\mathrm{hid}}$ to the channel error. For n independent hidden $\lambda\mathcal{D}[Z_i]$ terms, $\nu_{\mathrm{hid}} \le 2n\lambda$. Fixed hidden noise with all other resources fixed is therefore not covered by an arbitrarily scalable full-register fidelity claim. This agrees with the preserved acquisition converse.

Known-Hamiltonian accuracy. An integrated mismatch $\int_0^T \left\| H(t) - \widehat{H}(t) \right\| dt$ contributes at most twice that amount in channel distance, with the same actual ports. Acquiring an unknown many-body Hamiltonian from unknown inputs is a separate problem. The theorem uses known calibrated descriptions.

Loss. Perfect coherent delivery is an ideal premise. For uniform collection efficiency η_f on the Weyl outputs, their state-independent photon count gives an explicit sufficient comparison bound

$$\eta_{\mathrm{loss}} \le 2\sqrt{2\left(1 - e^{-R_{\mathrm{ph}}T(1-\sqrt{\eta_f})}\right)} \le 2\sqrt{2R_{\mathrm{ph}}T(1-\sqrt{\eta_f})}. \tag{50}$$

Indeed, the overlap of the lossless and beam-splitter-loss dilations is the scalar Poisson generating function $\mathbb{E}\eta_f^{M/2}$. A fixed nonzero loss is not thereby shown harmless as n grows; a sufficient loss budget scales with the required total fidelity. This is not a competitive lossy-device certificate.

Physical implementation level. The supplied implementation uses finite-dimensional receiver couplings to calibrated Markov fields, finite strength and slew, and finite endpoint gate time. These are the inherited control-model resources. They do not bound a microscopic system-field Hamiltonian or prove that a chosen fixed laboratory bandwidth supports the asymptotic family. Physical gate rates, memory protection, isolation, phase calibration, propagation and interface fidelity must meet the explicit budgets.

What is acquired. The recovered input can be used in any subsequent quantum experiment. Arbitrary discarded output states and their purifiers are not reconstructed. The mixed-output companion is not bypassed by claiming cheap whole-output storage. The pre-existing interaction-order theorem remains at its summed-term and full-retained-instrument scope; its weak-control proof boundary is unchanged.

11 Strongest ordinary comparators and scientific increment

Direct source exchange. If direct parallel SWAP into protected memory is actually available, it is the simpler baseline. This construction grants no such source interaction. Deliberately applied reversal gates act only on receiver memory.

Ordinary same-instrument collection without interactions. For one-site Hamiltonians, undoing the known rotations gives the exact product channel

$$\bigotimes_i \left[(1-e^{-rT})\,\mathrm{id} + e^{-rT}\mathcal{R}_{I/q}\right]. \tag{51}$$

Its diamond error is $2\{1-[1-(1-q^{-2})e^{-rT}]^n\}$. Ordinary first-reset collection therefore already has logarithmic waiting in this instrument. The new general-Hamiltonian theorem retains logarithmic collection time with an explicit interaction margin and an efficiently executable receiver decoder. It does not establish optimality among every possible coherent field decoder.

Petz and reversal. Inverse-dilation implementations, tabletop reversibility and continuous Petz dynamics are prior machinery [3, 4, 6]. General Petz algorithms are also available [5]. The special maximally mixed bath here removes the need for a density-matrix inverse or amplitude amplification, but that algebraic fact is not advertised as a new Petz theorem.

Operator size and classical high-noise simulation. Operator support/size methods in open systems are established [7]. Percolation-based simulation of noisy circuits is also prior machinery [8]. Neither by itself outputs an arbitrary quantum state entangled with an external reference. The scientific increment is the uniform support estimate attached to the explicit quantum recovery operation, together with the finite local acquisition of its inputs from the actual interacting output.

Global canonical collection. N1732 already supplies a finite general receiver. Applied globally, its d^2 levels correspond to $2n\log q$ qubits, but its dense matrix synthesis costs exponentially in n. The present receiver may store more qudits, yet its controls and compilation are local and polynomial. The gain is in an acquired scalable operation, not in confusing Hilbert-space dimension with qubit memory.

The precise historical novelty of this complete synthesis remains unestablished. The result is an asymptotic constructive acquisition theorem, not a demonstrated civilisation-changing device. It identifies a concrete regime in which a many-body experiment's coherent output can serve as a usable quantum input to subsequent computation.

12 Validation and preserved research

The main proof and normalization were independently audited. Finite exact-matrix calculations reconstruct the export channel and reverse-unitary decoder, compare their product with the operator-space projection identity, test the Lyapunov inequality, and reconstruct positive trace-preserving Choi matrices for a continuously interacting two-qubit example. The recorded identity discrepancies are below 6×10^{-16}. These are floating-point checks, not formal certificates or replacements for the proof.

For $H = X \otimes X + 0.37Z \otimes I + 0.23I \otimes Z$ and $r = 6$, the supplied decomposition gives $\beta = 2$ and $\kappa = 4$. Direct evolution of the finite operator-space recovery equation gives:

T	Computed $\left\|I - \overline{\mathcal{R}}_T\right\|_{2\to 2}$	Analytical $2e^{-4T}$
0.2	5.17357×10^{-1}	8.98658×10^{-1}
0.5	1.10668×10^{-1}	2.70671×10^{-1}
1.0	7.69868×10^{-3}	3.66313×10^{-2}
2.0	3.10336×10^{-5}	6.70925×10^{-4}

The table concerns the Hilbert–Schmidt residual in the proof; it is not a measured physical-device error or an independently computed diamond norm. The source code and unrounded values are included for reproduction.

The package includes the full independent construction and interface audit, source code and numerical results, a continuation handoff, and the unchanged previous research package. The exact digital causal-cut theorem, its ideal unitary-jump corollary, the mixed-output expander companion, and the interaction-order theorem retain their separate statements. The stationary-mirror-absorber route was examined and does not supply the missing arbitrary-input initialization operation by a dark-state argument alone.

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